

Numerical Quasi-Trefftz Method for Helmholtz problems

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 - Abderrahmane Bendali (IMT)
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- **Prototyping activity within the PEPR exascale Numpex (CNRS-INRIA-CEA)**
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 - Marc Fuentes (INRIA-SED)

Outline

1. Introduction
2. Classical variational Trefftz methods
3. Extension to general hyperbolic systems
4. Numerical quasi-Trefftz methods
5. Solving the large-scale linear system

Section 1

Introduction

Maxwell system

Find $\mathbf{E} \in L^2(\text{rot}, \Omega)$ and $\mathbf{H} \in L^2(\text{rot}, \Omega)$ such that

$$\begin{cases} \nabla \times \mathbf{H} &= ik \varepsilon_r \mathbf{E} \\ \nabla \times \mathbf{E} &= -ik \mu_r \mathbf{H} \text{ in } \Omega, \end{cases}$$

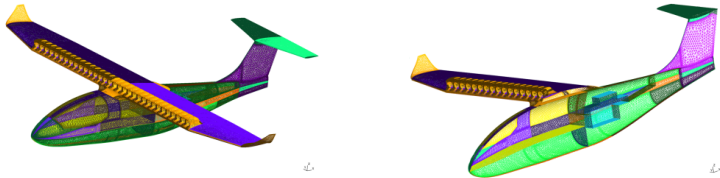
with boundary conditions

$$n \times \mathbf{H} - Z n \times (n \times \mathbf{E}) = \varphi \text{ on } \partial\Omega.$$

where Ω is a 3D domain, ε_0 and μ_0 are piecewise constant.

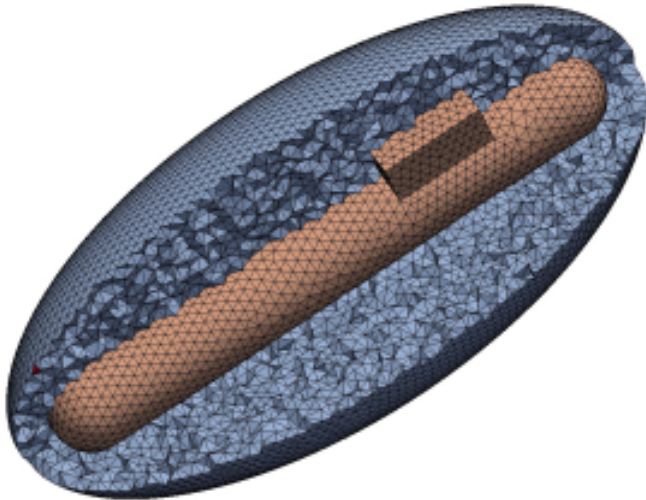
Objective: performing simulation in very large domains

→ 200 wavelengths = $200\lambda = 64 \times 10^6$ elements.



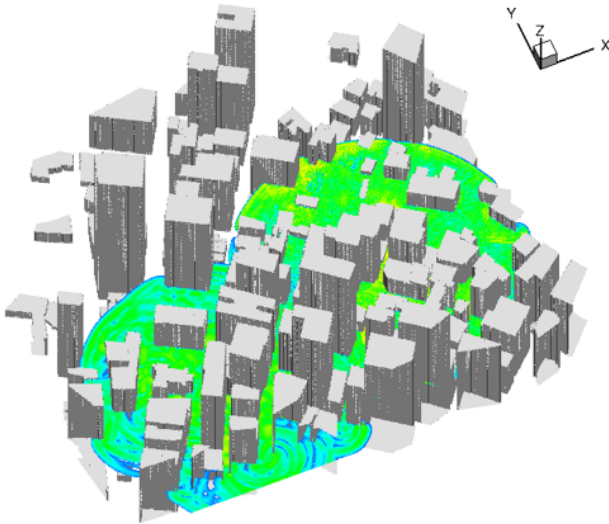
Geometry of Ampere prototype (DEMR ONERA).

Other examples of applications



Scattering by a submarine

Other examples of applications



Mesh of Manhattan by T. Volpert (ONERA)

Helmholtz equation

Helmholtz problem reads

$$\begin{cases} \Delta u + k^2 u = 0 & \text{in } \Omega, \\ \frac{\partial u}{\partial n} - iku = \varphi & \text{on } \partial\Omega. \end{cases}$$

Find $u \in H^1(\Omega)$ such that for any $v \in H^1(\Omega)$

$$\int_{\Omega} \nabla u \cdot \overline{\nabla v} - k^2 u \bar{v} - ik \int_{\partial\Omega} u \bar{v} = \int_{\partial\Omega} \varphi \bar{v}$$

From this **weakly elliptic** form, we easily get

- the well-posedness (Fredholm alternative, unique continuation)
- **convergent** finite element formulations

Is it the end of the story ?

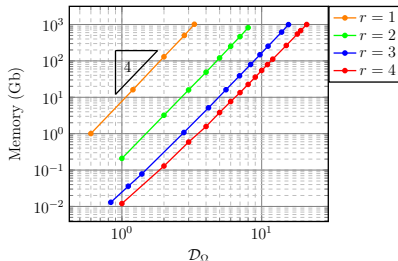
F. Ihlenburg and I. Babuska, *Finite element solution of the Helmholtz equation with high wave number part II: the hp version of the FEM*, *SIAM Journal on Numerical Analysis*, 34 (1), pp. 315–358 (1997).

Oliver G. Ernst, and Martin J. Gander. "Why it is difficult to solve Helmholtz problems with classical iterative methods." *Numerical analysis of multiscale problems* (2012): 325-363.

Memory limits of classical solvers

$$\mathbf{A} \mathbf{x} = \mathbf{F}$$

- A minimal number of unknowns:
 - second-order PDE
 - continuous finite elements
- Finite element methods handle geometry and evanescent modes very well.
- **Memory limitation** for LU methods.



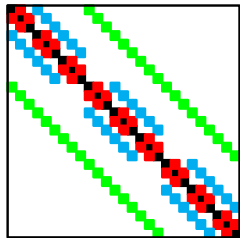
Memory cost (factorization LU of MUMPS[®]) for different sizes of domain $\mathcal{D}_\Omega$ and different approximation orders r .

$$\text{Memory} = (\mathcal{D}_\Omega)^4.$$

	Memory	Sockets	$\mathcal{D}_\Omega$	Power	Cost
PC	32 GB	1	10 λ	200 W	0.034 €/h
Parallel computing	1 TB	16	24 λ	3.2 kW	0.544 €/h
HPC	320 TB	5000	100 λ	1.0 MW	170 €/h
HPC+	5120 TB	80000	200 λ	16 MW	2720 €/h

Matrix structure

$$AX = F$$



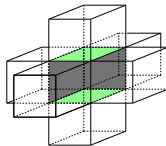
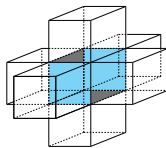
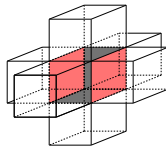
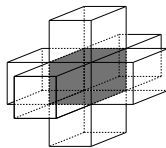
A

Matrix A for $27 = 3^3$ cubes.

X

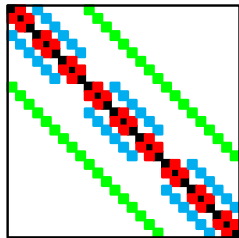
$=$

F



Matrix structure

$$AX = F$$



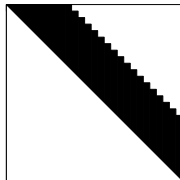
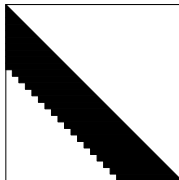
A

Matrix A for $27 = 3^3$ cubes.

$$\begin{array}{c} \text{Yellow bar} \\ X \end{array} = \begin{array}{c} \text{Yellow bar} \\ F \end{array}$$

$$A = LU$$

- L : lower triangular matrix
- U : upper triangular matrix



	L				U
Size of the domain	3λ	5λ	10λ	20λ	100λ
Number of elements	27	512	1000	8000	10^6
LU Memory	10 MB	135 MB	4.3 GB	138 GB	432 TB

There exist partial solutions

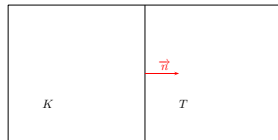
– Global preconditioners

- A. Vion and C. Geuzaine. Double sweep preconditioner for optimized Schwarz methods applied to the Helmholtz problem. J. Comput. Phys., 2014.
- M. Gander, L. Halpern, KS. L. Repiquet. A new coarse grid correction for RAS/AS. In Domain Decomposition Methods in Science and Engineering XXI 2014.

– Domain decomposition methods

$$\begin{cases} Y p_T^{k+1} + i\kappa \nabla p_T^{k+1} \cdot \vec{n} = Y p_K^k + i\kappa \nabla p_K^k \cdot \vec{n}, \\ Y p_K^{k+1} - i\kappa \nabla p_K^{k+1} \cdot \vec{n} = Y p_T^k - i\kappa \nabla p_T^k \cdot \vec{n}, \end{cases}$$

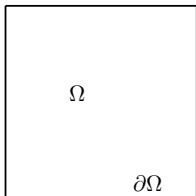
- A. Modave, X. Antoine et C. Geuzaine. An efficient domain decomposition method with cross-point treatment for Helmholtz problems. In CSMA 2019-14e Colloque National en Calcul des Structures. 2019.
 - F. Collino, P. Joly, and M. Lecouvez. "Exponentially convergent non overlapping domain decomposition methods for the Helmholtz equation." ESAIM: Math. Mod. and Num. An. 54.3. 775-810. 2020.
- ## – Hyperbolic Trefftz methods
- Back to hyperbolicity
 - Naturally adapted to GMRES methods



Section 2

Classical variational Trefftz methods

Classical variational Trefftz methods



The sample problem reads: find $p \in H^1(\Omega)$ such that

$$\begin{cases} \Delta p + k^2 p = 0 & \text{in } \Omega, \\ \frac{\partial p}{\partial n} - ikp = \varphi & \text{on } \partial\Omega. \end{cases}$$

with $\varphi \in L^2(\partial\Omega)$.

Main features of the Trefftz space are

- it belongs to the discontinuous Galerkin spaces
- it is composed of local solutions to the PDE of interest

It is denoted by X and defined by

$$X = \left\{ p' : \Omega \longrightarrow \mathbb{C} \text{ such that } p'_K \in X_K \right\}$$

p'_K is the restriction of p' to element K and

$$X_K = \left\{ p'_K \in H^1(K) \text{ such that } \Delta p'_K + k^2 p'_K = 0, \quad \frac{\partial p'_K}{\partial n} \in L^2(\partial K) \right\}$$

Mesh of the domain Ω

- $K \in \mathcal{K}$ denotes one element of the mesh, ∂K its boundary

- $\mathcal{K}$ is the set of elements and $\bar{\Omega} = \bigcup_{K \in \mathcal{K}} \bar{K}$.

$$\partial K = \bigcup_{F \in \mathcal{F}_K} F$$

- $\mathcal{F}_K$ denotes the set of the faces of K

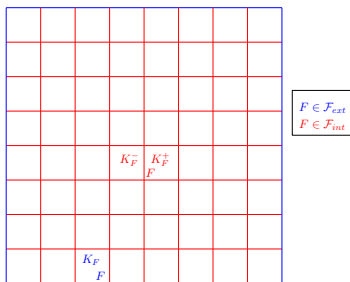
- F stands for a face

- $\mathcal{F}_{int}$ is the set of **interior faces**

- $\mathcal{F}_{ext}$ is the set of **exterior faces**

- If F is an **interior face**, then it hits two elements K_F^+ and K_F^- .

- If F is an **exterior face**, then it touches only one element K_F .



Reciprocity principle

Green formula reads:

$$\int_{\partial K} \frac{\partial p_K}{\partial n} \overline{p'_K} = \int_K \nabla p_K \cdot \overline{\nabla p'_K} - k^2 p_K \overline{p'_K} = \int_{\partial K} p_K \overline{\frac{\partial p'_K}{\partial n}} \quad \text{for } p' \in X.$$

We sum up all the elements:

$$\sum_{K \in \mathcal{K}} \int_{\partial K} \nabla p_K \cdot \vec{n}_K \overline{p'_K} - p_K \overline{\nabla p'_K \cdot \vec{n}_K} = 0,$$

We rewrite this expression in terms of jumps between elements:

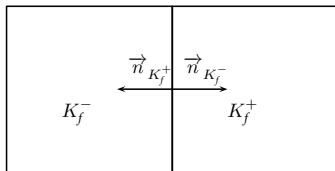
$$\sum_{F \in \mathcal{F}_{int}} \int_F \nabla p_F \cdot \overline{[p']_F} - p_F \overline{[\nabla p']_F} + \sum_{F \in \mathcal{F}_{ext}} \int_F \nabla p_F \cdot \vec{n}_{K_F} \overline{p'_{K_F}} - p_F \overline{\nabla p'_{K_F} \cdot \vec{n}_{K_F}} = 0,$$

assuming the solution is continuous on F

- p_F and $\vec{v}_F$ are the respective trace p and $\vec{v}$ on F ,
- $\vec{n}_F$ is the outgoing normal vector to the exterior face F .

The **jumps** are defined by

$$\begin{cases} [\vec{v}]_F = \vec{v}_{K_F^+} \cdot \vec{n}_{K_F^+} + \vec{v}_{K_F^-} \cdot \vec{n}_{K_F^-}, \\ [p]_F = p_{K_F^+} \vec{n}_{K_F^+} + p_{K_F^-} \vec{n}_{K_F^-}, \end{cases}$$



Numerical traces and the choice of $\alpha, \beta, \gamma, \delta$

Numerical traces approximate ∇p_F and p_F (stamped by a *hat*):

$$\sum_{F \in \mathcal{F}_{int}} \int_F \widehat{\nabla p_F} \cdot \overline{[p']_F - \widehat{p_F}} \overline{[\nabla p']_F} + \sum_{F \in \mathcal{F}_{ext}} \int_F \widehat{\nabla p_F} \cdot \vec{n}_{K_F} \overline{p'_{K_F} - \widehat{p_F}} \overline{\nabla p'_{K_F} \cdot \vec{n}_{K_F}} = 0,$$

$$\widehat{\nabla p_F} = \frac{\nabla p_{K^+} + \nabla p_{K^-}}{2} + \gamma [p]_F + \delta [\nabla p]_F \quad \text{and} \quad \widehat{p_F} = \frac{p_{K^+} + p_{K^-}}{2} + \alpha [p]_F + \beta [\nabla p]_F$$

Objective: get **coercive** formulations adapted to **iterative solvers**.

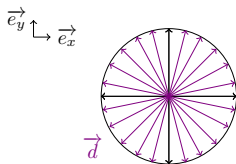
- Arnold's IPDG formulation: $\alpha = \beta = \delta = 0$, $\gamma = \gamma_0/h^2$. **Coercive**, but **it does not go through an iterative process**.
- **Outgoing flux method:** with the ingoing flux $\nabla p \cdot n - ikp$ and the outgoing flux $\nabla p \cdot n + ikp$, the numerical trace is a combination of the two *outgoing* traces

$$\widehat{p_F} = \frac{1}{2ik} \left(\nabla p_{K_F^+} \cdot n_{K_F^+} + ikp_{K_F^+} \right) + \frac{1}{2ik} \left(\nabla p_{K_F^-} \cdot n_{K_F^-} + ikp_{K_F^-} \right),$$

i.e. $\alpha = 0$, $\beta = \frac{1}{2ik}$,

and likewise for $\widehat{\nabla p_F}$. The formulation is now **coercive and iterative** – but this construction is **difficult to generalize** to more complex equations.

The classical discrete Trefftz space



The basis functions of the discrete space are:

$$w_{d_n}(\mathbf{x}) = \exp\left(ikd_n \cdot \mathbf{x}\right),$$

with

- d_n , the direction of the plane wave
- $k = \frac{\omega}{c}$, the wave number

The basis functions are discontinuous:

$$p_h(\mathbf{x}) = \sum_{i=1}^I [p_h]_{K,i} w_{d_i}(\mathbf{x})(\mathbf{x})$$



This discretization leads to

- easy-to-perform analytical calculations
- a **very ill-conditioned** linear system.

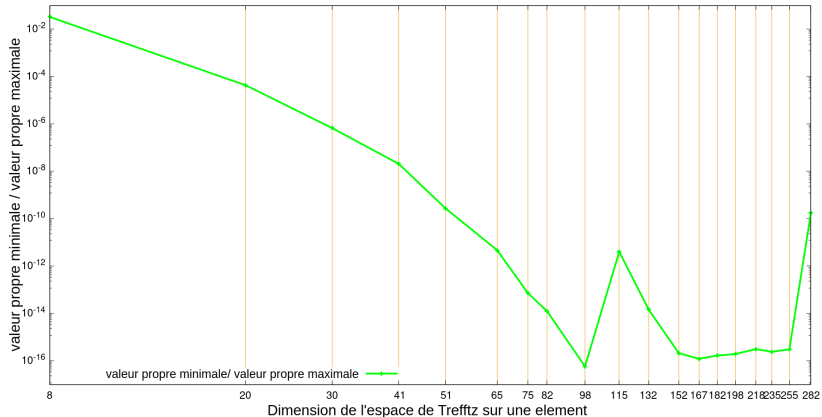
Conditioning problem

Theoretically, plane waves form a **free family**.

$$w_{d_n}(\mathbf{x}) = \exp(ikd_n \cdot \mathbf{x}),$$

Numerically, this family is almost linked.

$$R_{m,m'} = \int_K w_{d_m}(\mathbf{x}) \overline{w_{d_{m'}}(\mathbf{x})}$$



Ratio of the smallest to the largest eigenvalue of R versus the dimension of the local Trefftz space.

Section 3

Extension to general hyperbolic systems

Revisiting Trefftz methods with the hyperbolic formalism

Wave propagation problems are

- **quasi-elliptic** at the **local** level.

$$\Delta p + \omega^2 p = 0$$

- **hyperbolic** over long distances.

$$\left\{ \begin{array}{l} -i\omega \mathbf{v} = \nabla p \\ -i\omega p = \operatorname{div}(\mathbf{v}) \end{array} \right. \iff \left\{ \begin{array}{ll} \frac{\partial \mathbf{v}}{\partial t} = \nabla p & \text{with } p(x, t) = p(x) \exp(-i\omega t), \\ \frac{\partial p}{\partial t} = \operatorname{div}(\mathbf{v}) & \text{with } \mathbf{v}(x, t) = \mathbf{v}(x) \exp(-i\omega t), \end{array} \right.$$

Is it possible to better understanding wave problems by introducing Friedrichs' formalism?

- what is a Friedrichs' system?
- What basic information does it contain?
- How to understand Riemann's solvers in this framework?
- What boundary conditions?

Friedrichs' systems

Definition of symmetrical Friedrichs systems

Find $Y : \Omega \times \mathbb{R}^+ \longrightarrow \mathbb{R}^m$ with $\Omega \subset \mathbb{R}^d$

$$M(\mathbf{x}) \frac{\partial Y}{\partial t}(\mathbf{x}, t) + \sum_{j=1}^d \frac{\partial}{\partial x_j} F_j(Y)(\mathbf{x}, t) = 0$$

- $M : \Omega \longrightarrow \mathbb{R}^{m \times m}$ is a field of positive definite symmetric matrices
- The symbol for the differentiation operator in space

$$F : \mathbb{R}^d \longrightarrow \mathbb{R}^{m \times m} \quad \text{with} \quad F[\xi] = \sum_{j=1}^d \xi_j F_j$$

with $F_j \in \mathbb{R}^{m \times m}$, $1 \leq j \leq d$, symmetrical matrices.

Theorem: Energy conservation formula

$$\frac{\partial}{\partial t} \int_{\Omega} \frac{Y^T(\mathbf{x}, t) M(\mathbf{x}) Y(\mathbf{x}, t)}{2} d\mathbf{x} + \frac{1}{2} \int_{\partial\Omega} Y^T(\mathbf{x}, t) F[\vec{n}] Y(\mathbf{x}, t) = 0$$

- First term: total energy
- Second term: work of external forces

First example of Friedrichs system: acoustics in 3D

$$\left. \begin{aligned} \rho \frac{\partial \mathbf{v}}{\partial t}(\mathbf{x}, t) + \nabla p(\mathbf{x}, t) &= 0, \\ \chi \frac{\partial p}{\partial t}(\mathbf{x}, t) + \operatorname{div} \mathbf{v}(\mathbf{x}, t) &= 0. \end{aligned} \right\} \iff \begin{cases} \rho \frac{\partial \mathbf{v}_1}{\partial t}(\mathbf{x}, t) + \frac{\partial p}{\partial x_1}(\mathbf{x}, t) = 0, \\ \rho \frac{\partial \mathbf{v}_2}{\partial t}(\mathbf{x}, t) + \frac{\partial p}{\partial x_2}(\mathbf{x}, t) = 0, \\ \rho \frac{\partial \mathbf{v}_3}{\partial t}(\mathbf{x}, t) + \frac{\partial p}{\partial x_3}(\mathbf{x}, t) = 0, \\ \chi \frac{\partial p}{\partial t}(\mathbf{x}, t) + \frac{\partial \mathbf{v}_1}{\partial x_1}(\mathbf{x}, t) + \frac{\partial \mathbf{v}_2}{\partial x_2}(\mathbf{x}, t) + \frac{\partial \mathbf{v}_3}{\partial x_3}(\mathbf{x}, t) = 0. \end{cases}$$

The unknown and the hyperbolic matrices

$$Y = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ p \end{bmatrix}, \quad F_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \quad F_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad F_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The hyperbolic symbol

$$M = \begin{bmatrix} \rho & 0 & 0 & 0 \\ 0 & \rho & 0 & 0 \\ 0 & 0 & \rho & 0 \\ 0 & 0 & 0 & \chi \end{bmatrix}, \quad F[\xi] = \begin{bmatrix} 0 & 0 & 0 & \xi_1 \\ 0 & 0 & 0 & \xi_2 \\ 0 & 0 & 0 & \xi_3 \\ \xi_1 & \xi_2 & \xi_3 & 0 \end{bmatrix}$$

- Total energy of the system: $\int_{\Omega} \frac{Y^T M Y}{2} = \int_{\Omega} \frac{\rho |\mathbf{v}|^2 + \chi p^2}{2} d\mathbf{x}$
- Work of external forces: $\int_{\partial\Omega} \frac{Y^T F[n] Y}{2} = \int_{\partial\Omega} p \mathbf{v} \cdot \mathbf{n} ds_{\mathbf{x}}$

Second example of Friedrichs' system: Maxwell system in 3D

$$\begin{cases} \varepsilon_0(\mathbf{x}) \frac{\partial \mathbf{E}}{\partial t} - \operatorname{rot} \mathbf{H} = 0, \\ \mu_0(\mathbf{x}) \frac{\partial \mathbf{H}}{\partial t} + \operatorname{rot} \mathbf{E} = 0. \end{cases}$$

The unknown and the hyperbolic matrices

$$Y = \begin{bmatrix} E_1 \\ E_2 \\ E_3 \\ H_1 \\ H_2 \\ H_3 \end{bmatrix}, \quad F_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad F_2 = \dots, \quad F_3 = \dots$$

$$M = \begin{bmatrix} \varepsilon_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \varepsilon_0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \varepsilon_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu_0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu_0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu_0 \end{bmatrix}, \quad F[\xi] = \begin{bmatrix} 0 & 0 & 0 & 0 & \xi_3 & -\xi_2 \\ 0 & 0 & 0 & -\xi_3 & 0 & \xi_1 \\ 0 & 0 & 0 & \xi_2 & -\xi_1 & 0 \\ 0 & -\xi_3 & \xi_2 & 0 & 0 & 0 \\ \xi_3 & 0 & -\xi_1 & 0 & 0 & 0 \\ -\xi_2 & \xi_1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- Total energy of the system: $\int_{\Omega} \frac{Y^T M Y}{2} = \int_{\Omega} \frac{\varepsilon_0 |\mathbf{E}|^2 + \mu_0 |\mathbf{H}|^2}{2} d\mathbf{x}$
- Work of external forces: $\int_{\partial\Omega} \frac{Y^T F[n] Y}{2} = - \int_{\partial\Omega} (\mathbf{E} \times \mathbf{H}) \cdot \mathbf{n} ds_{\mathbf{x}}$

Hyperbolic problems in frequency regime

We seek a solution with a harmonic time dependency:

$$Y(\mathbf{x}, t) = \Re(Y(\mathbf{x}) \exp(-i\omega t))$$

Find $Y \in L^2(\Omega)$ such that

$$\begin{cases} -i\omega M(\mathbf{x})Y(\mathbf{x}) + \sum_{i=1}^n \frac{\partial}{\partial x_i} F_i Y(\mathbf{x}) = 0, & \mathbf{x} \in \Omega, \\ F_-[n]Y(\mathbf{x}) = \varphi(\mathbf{x}), & \mathbf{x} \in \partial\Omega, \end{cases}$$

The variational space is:

$$H = \left\{ Y \in (L^2(\Omega))^k \text{ such that } \sum_{i=1}^d \frac{\partial F_i Y}{\partial x_i} \in (L^2(\Omega))^k \right\}$$

The hyperbolic problem can be interpreted as a conservative system introducing the divergence operator

$$\operatorname{div}(FY) = \sum_{i=1}^n \frac{\partial}{\partial x_i} F_i Y(\mathbf{x})$$

We have

$$\begin{cases} -i\omega M(\mathbf{x})Y(\mathbf{x}) + \operatorname{div}(FY) = 0, & \mathbf{x} \in \Omega, \\ F_-[n]Y(\mathbf{x}) = \varphi(\mathbf{x}), & \mathbf{x} \in \partial\Omega, \end{cases}$$

General plane waves solutions

$$Y(\mathbf{x}) = Y_0 \exp(i\mathbf{k} \cdot \mathbf{x}) \quad \text{on } \mathbb{R}^d$$

- The wave vector $\mathbf{k}$
- The direction of the wave

$$\mathbf{k} = \pm |\mathbf{k}| \mathbf{d}$$

- The amplitude of the wave Y_0
- The amplitude Y_0 is an eigenvector of the hyperbolic symbol $F[\mathbf{d}]$ associated to the eigenvalue $\frac{\omega}{|\mathbf{k}|}$

$$\pm |\mathbf{k}| F[\mathbf{d}] Y_0 = \omega M Y_0$$

Solve the well-posed eigenvalue problem :

The direction $\mathbf{d}$ given, find $Y_0 \in \mathbb{R}^m$ and $\lambda \in \mathbb{R}$ such that

$$F[\mathbf{d}] Y_0 = \lambda M Y_0 \quad \Longleftrightarrow \quad M^{-\frac{1}{2}} F[\mathbf{d}] M^{-\frac{1}{2}} (M^{\frac{1}{2}} Y_0) = \lambda (M^{\frac{1}{2}} Y_0).$$

Y_0 describes the polarization of the wave : transverse, longitudinal, or mixed.

Dispersion relation :

$$\det(-\omega M + F[\mathbf{k}]) = 0$$

Incoming and outgoing waves

The direction $\mathbf{d}$ being given, we have

$$F[\mathbf{d}] = MP[\mathbf{d}]\Lambda[\mathbf{d}]P^T[\mathbf{d}]M$$

We associate to the hyperbolic symbol $F[n]$ a spectral **decomposition** in positive and negative eigenspaces

$$F[n] = F_+[n] + F_-[n]$$

with $F_+[n]$ a positive symmetric matrix and $F_-[n]$ a negative symmetric matrix

$$F_+[n] = MP[\mathbf{d}]\Lambda_+[\mathbf{d}]P^T[\mathbf{d}]M \text{ and } F_-[n] = MP[\mathbf{d}]\Lambda_-[\mathbf{d}]P^T[\mathbf{d}]M$$

- The positive part is associated to outgoing waves
- the negative part is associated to incoming waves

The hyperbolic system is closed by a boundary condition that prescribes the incoming waves

$$F_-[n]Y = \varphi \text{ on } \partial\Omega$$

which models the information coming from the exterior of the domain.

This is a concept of **outgoing radiation conditions** generalizing Sommerfeld's radiation condition.

The Trefftz space and the reciprocity principle

The **hyperbolic integration by parts** reads, for all Y and $Y' \in H$,

$$\int_{\Omega} \operatorname{div} (FY) \cdot \overline{Y'} + \int_{\Omega} Y \cdot \operatorname{div} (\overline{FY'}) = \int_{\partial\Omega} F[n]Y \cdot \overline{Y'}.$$

If Y and Y' both solve $-i\omega MY + \operatorname{div} (FY) = 0$, the two volume terms cancel and

$$\int_{\partial K} \overline{Y'}^T F[\mathbf{n}]Y = 0 : \quad \text{the **reciprocity principle**.}$$

The Trefftz space is the space of functions solving the hyperbolic problem in each element K :

$$W = \prod_{K \in \mathcal{K}} \left\{ Y_K \in (L^2(K))^k \mid \operatorname{div} (FY_K) \in (L^2(K))^k \text{ and } F[n]Y_K \in L^2(\partial K) \right\}$$

Both the solution and the test functions are taken in W ; the extra regularity is exactly what makes the flux $F[n]Y_K$ well defined on each face. Summing over the mesh,

$$\sum_{K \in \mathcal{K}} \int_{\partial K} F[\mathbf{n}]Y \cdot \overline{Y'_K} = 0.$$

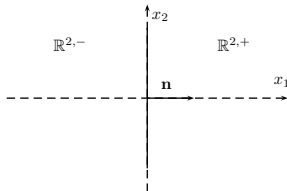
- The exact solution is continuous across the faces, so $F[n]Y$ is well defined there.
- The test functions are *not*: $F[n]Y'$ is two-valued on a face — hence the need for a **numerical flux**.

Interior Riemann flux

Solve the acoustic problem set in $\mathbb{R}^d$

- For $x_1 < 0$, we have

$$\begin{cases} A_- \frac{\partial \vec{v}}{\partial t}(\mathbf{x}, t) + \nabla p(\mathbf{x}, t) = 0, \\ \chi_- \frac{\partial p}{\partial t}(\mathbf{x}, t) + \operatorname{div} \vec{v}(\mathbf{x}, t) = 0, \end{cases}$$



- For $x_1 > 0$, we have

$$\begin{cases} A_+ \frac{\partial \vec{v}}{\partial t}(\mathbf{x}, t) + \nabla p(\mathbf{x}, t) = 0, \\ \chi_+ \frac{\partial p}{\partial t}(\mathbf{x}, t) + \operatorname{div} \vec{v}(\mathbf{x}, t) = 0, \end{cases}$$

We then obtain

with initial conditions

$$\left\{ \begin{array}{ll} p(\mathbf{x}, 0) = p_- & \text{for } x_1 < 0, \\ p(\mathbf{x}, 0) = p_+ & \text{for } x_1 > 0, \\ \vec{v}(\mathbf{x}, 0) = \vec{v}_- & \text{for } x_1 < 0, \\ \vec{v}(\mathbf{x}, 0) = \vec{v}_+ & \text{for } x_1 > 0. \end{array} \right.$$

$$\hat{p}(0,t) = \frac{Y_{-p_-} + Y_{+p_+}}{Y_{-} + Y_{+}} + \frac{Y_{-}Y_{+}}{Y_{-} + Y_{+}} \llbracket \vec{v} \rrbracket_F,$$

$$\hat{v}(0, t) = \frac{Z_- v_- + Z_+ v_+}{Z_- + Z_+} + \frac{Z_- Z_+}{Z_- + Z_+} \llbracket p \rrbracket_F$$

with admittance Y and impedance Z :

and the transmission conditions in $x_1 = 0$:

$$Y_{\pm} = \frac{1}{Z_{\pm}} = \sqrt{\frac{n \cdot A_{\pm} n}{\chi_{\pm}}}$$

$$\|p\|_F = 0 \quad \text{and} \quad \|\vec{v}\|_F = 0$$

Conclusion for the section

- We have shown how to get the interior Riemann flux

$$\widehat{F[n]Y} = F[n] \frac{Y_{K_F^+} + Y_{K_F^-}}{2} + T[\![Y]\!]_f$$

- It is possible to do the same with the boundary condition

$$\widehat{F[n]Y} = F[n]Y - \left(F[n]_- Y - \varphi \right) = F[n]_+ Y + \varphi$$

We end up with a Trefftz problem

$$\sum_{f \in \mathcal{F}_{int}} \int_f \widehat{F[n]Y}_f \cdot \overline{\llbracket Y' \rrbracket_F} + \sum_{f \in \mathcal{F}_{ext}} \int_f \widehat{F[n]Y}_f \cdot \overline{Y'_K} = 0$$

This formulation is **coercive** ! It can be solved iteratively by a block Jacobi algorithm.

This is really surprising since the original problem is not coercive. For example, the Helmholtz problem reads

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla \bar{v} - k^2 \int_{\Omega} u \bar{v} - ik \int_{\partial\Omega} u \bar{v}$$

Section 4

Numerical quasi-Trefftz methods

The problem of round-off errors

In 3D, **the theory** tells us that Trefftz methods are

- convergent
- well-posed

In **practice**

- Occasionally non-invertible
- Often non-convergent

Why?

- Two plane wave functions related to closely parallel directions are numerically linked.

Our objective: reduce the number of basis functions by eliminating those that are unnecessary

- Decrease the computational costs
- Reduce the **round-off errors**

The approach of this talk : Quasi-Trefftz method¹

Other approaches to this redundancy:

- generalised plane waves and polynomial quasi-Trefftz functions (Imbert-Gérard)²
- evanescent plane waves (Moiola)³
- embedded Trefftz DG (Lehrenfeld)⁴

¹M. Rivet, S. Pernet, S. Tordeux, 'Optimised Correction Polynomial Functions for the Flux Reconstruction Method', Appl. Math. Lett., 2024.

²L.-M. Imbert-Gérard, B. Després, 'A generalized plane wave numerical method', IMA J. Numer. Anal. 34, 2014; L.-M. Imbert-Gérard, G. Sylvand, 'Three types of quasi-Trefftz functions', IMA J. Numer. Anal. 45(4), 2025.

³E. Parolin, D. Huybrechs, A. Moiola, 'Stable approximation of Helmholtz solutions by evanescent plane waves', ESAIM: M2AN 57(6), 2023.

⁴C. Lehrenfeld, P. Stocker, 'Embedded Trefftz discontinuous Galerkin methods', Int. J. Numer. Methods Eng. 124(17), 2023.

Basis reduction strategy (1): the Gram matrix and its SVD

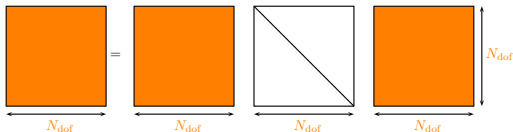
A basis reduction

- does not consist in eliminating basis functions,
- consists in keeping a few well-chosen **linear combinations** of them.

The scalar product and the discrete Galerkin basis define a Gram matrix

$$M = (\mathbf{w}_i^K, \mathbf{w}_j^K)_K \quad \text{with } 1 \leq i, j \leq N_{\text{dof}}$$

which can be diagonalized (SVD decomposition)¹



$$M = P\Lambda P^*$$

with P an orthogonal matrix and Λ a positive diagonal matrix. We keep only the N_{red} largest eigenvalues,

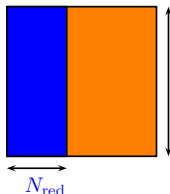
$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{N_{\text{red}}} \geq \varepsilon > \lambda_{N_{\text{red}}+1} \geq \cdots \geq \lambda_{N_{\text{dof}}},$$

Little information is lost, since the discarded eigenvalues are small.

¹S. Congreve, J. Gedicke, I. Perugia, 'Numerical Investigation of the Conditioning for Plane Wave Discontinuous Galerkin Methods', in Numerical Mathematics and Advanced Applications ENUMATH 2017, pp. 493–500, 2019, Springer, Cham.

Basis reduction strategy (2): the reduced basis

Truncating keeps only the N_{red} leading eigenvectors, which defines $\mathbf{P}_{red} =$



and $\mathbf{M} \simeq \mathbf{M}_{red} = \mathbf{P}_{red} \mathbf{\Lambda}_{red} \mathbf{P}_{red}^*$.

The **reduced basis** is composed of the combination of the original basis functions defined by the columns of $\mathbf{P}_{red}$.

$$w_{red}^i = \sum_{j=1}^{N_{dof}} p_{j,i}^{red} w^j \quad \text{with } 1 \leq i \leq N_{red} \text{ and } 1 \leq j \leq N_{dof}.$$

i.e. $(w_{red}^1, \dots, w_{red}^{N_{red}}) = (w^1, \dots, w^{N_{dof}}) \mathbf{P}_{red}$, so that

$$\mathbf{x} = \sum_{i=1}^{N_{dof}} w^i x_i = \sum_{j=1}^{N_{red}} w_{red}^j x_j^{red} \quad \text{and} \quad \mathbf{x} = \mathbf{P}_{red} \mathbf{x}^{red},$$

which defines the **reduced Galerkin problem**: find $Y_h^{red} \in W_h^{red}$ such that $a(Y_h^{red}, Y_h') = \ell(Y_h')$ for all $Y_h' \in W_h^{red}$.

The reduction of the basis improves the memory

N_{red} depends on the size of element K and on the truncation criterion ε .

Size of $K \backslash \varepsilon$	10^{-13}	10^{-11}	10^{-9}	10^{-7}	10^{-5}	10^{-4}	10^{-3}	10^{-2}
0.25λ	154	126	96	70	48	48	30	24
0.5λ	190	186	174	132	96	84	70	48
1λ	196	196	196	190	180	174	148	114

0.25λ case, $N_{\text{kry}}^{\text{restart}} = 500$.

10^{-15}	ε	10^{-13}	10^{-11}	10^{-9}	10^{-7}	10^{-5}	10^{-4}	10^{-3}	10^{-2}
N = 196	N_{red}	154	126	96	70	48	36	30	16
1.57 GB	Memory cost (GB)	1.23	1.01	0.76	0.56	0.38	0.28	0.24	0.13

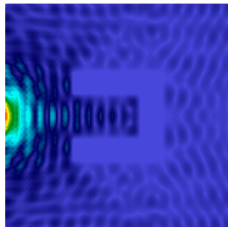
The iterative solution does not converge without any reduction
 $\Rightarrow$ Improvement of spectral properties

How far can we reduce the basis?

ε	10^{-7}	10^{-5}	10^{-4}	10^{-3}	10^{-2}
Error	N/A	3.4×10^{-2}	9.8×10^{-2}	0.15	0.74
Memory cost (GB)	0.56	0.38	0.28	0.24	0.13
Time (hours)	55	27	11.3	7.23	0.58

Too much reduction

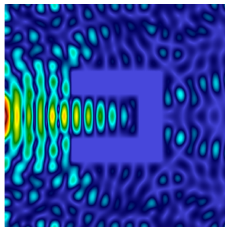
$$N_{\text{red}} = 16 \quad \varepsilon = 10^{-2}$$



Electromagnetic Field Magnitude
0.0e+00 4 6 8 1.1e+01

Good compromise

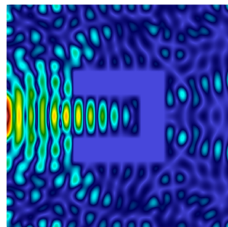
$$N_{\text{red}} = 36 \quad \varepsilon = 10^{-4}$$



Electromagnetic Field Magnitude
0.0e+00 4 6 8 1.1e+01

Reference solution

$$N_{\text{red}} = 70 \quad \varepsilon = 10^{-7}$$



Electromagnetic Field Magnitude
0.0e+00 4 6 8 1.1e+01

Quasi-Trefftz methods

- We introduce the solution operator

$$S_K : L^2(\partial K) \longrightarrow H^1(K) \quad \varphi \longmapsto p_K$$

$$\begin{cases} \Delta p_K + k^2 p_K = 0 & \text{in } K, \\ \nabla p_K \cdot n - ikp_K = \varphi & \text{on } \partial K, \end{cases}$$

along with its approximation $S_{K,h} : \varphi \longrightarrow p_{K,h}$ for instance

- finite elements
- boundary finite elements (Bendali, Barucq)
- spectral differences and FR (Matthias Rivet, Sébastien Pernet)
- A finite element subspace V_H of $L^2(\partial K)$ as well as the discrete Trefftz and quasi-Trefftz spaces

$$X_H = S_K V_H \quad \text{and} \quad X_{H,h} = S_{K,h} V_H$$

- A **master Trefftz solver** couples the different local solution methods, through the variational formulation of the previous section.

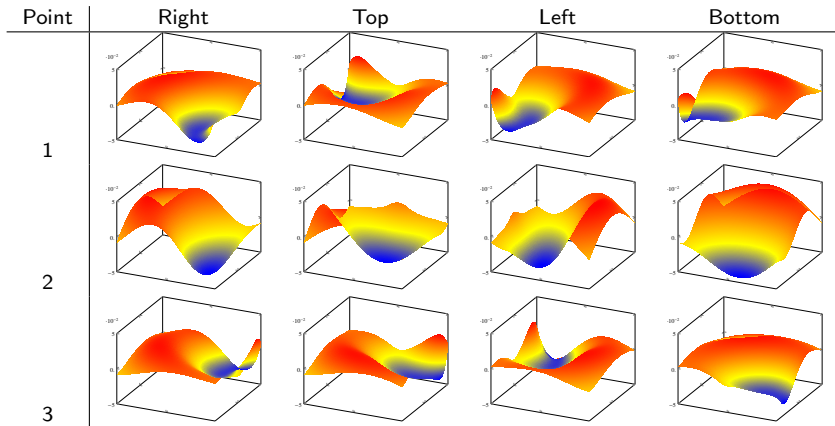
Computational cost

- A single LU factorization per element and several solutions.
- Adapted to MPI.

Local solution cost ($C N_{elem}$) $\ll$ cost of the master solver ($C' N_{elem}^{\frac{4}{3}}$)

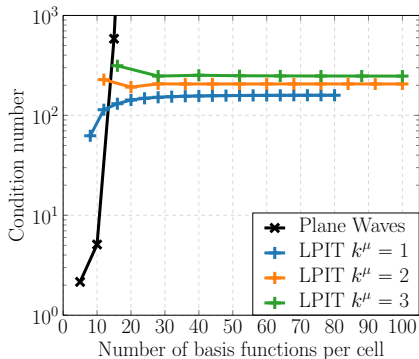
Examples of quasi-Trefftz basis functions

Piecewise polynomial boundary conditions and their associated quasi-Trefftz basis functions (real part):

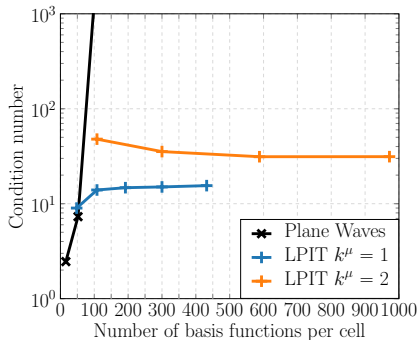


Each basis function is the local solution associated with a polynomial boundary condition localized at one point on one side of the cell (Matthias Rivet).

The quasi-Trefftz basis improves the conditioning



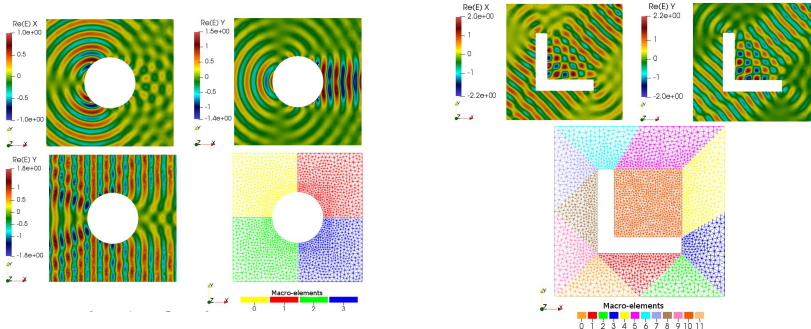
(a) 2D Helmholtz



(b) 3D Maxwell

- **Plane waves**: the condition number of the local mass matrix **explodes** with the number of basis functions per cell.
- **Quasi-Trefftz** basis: **controlled conditioning**, even for a large number of basis functions.

Two qualitative examples

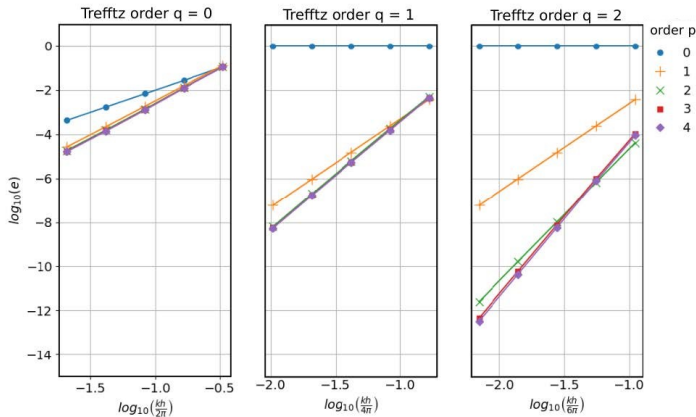


- Barucq, Bendali, Fares, Mattesi, Tordeux, A symmetric Trefftz-DG formulation based on a local boundary element method for the solution of the Helmholtz equation, JCP, (2017)
- Fure, Pernet, Sirdey, Tordeux, A discontinuous Galerkin Trefftz type method for solving the two dimensional Maxwell equations, PDE and Applications (2020)
- Rivet, Pernet, Tordeux, Optimised Correction Polynomial Functions for the Flux Reconstruction Method in Time-Harmonic Electromagnetism, Applied Mathematics Letters (2024)

Optimal convergence order

The order of the method is defined in two ways:

- the space order of the Trefftz method characterized by the trace space
- the order of the auxiliary method



Superconvergence of quasi-Trefftz methods: expected order $+\frac{1}{2}$

Section 5

Solving the large-scale linear system

Krylov methods and GMRES in practice

Cayley–Hamilton theorem

If $\mathbf{A} \in \mathbb{C}^{n \times n}$ is invertible, there exists a polynomial p with

$$\mathbf{A}^{-1} = p(\mathbf{A}), \quad \deg(p) < n.$$

Krylov method for $\mathbf{AX} = \mathbf{F}$

$$\mathbb{K}_k = \text{span}\left(\left\{\mathbf{F}, \mathbf{A}\mathbf{F}, \dots, \mathbf{A}^{k-1}\mathbf{F}\right\}\right)$$

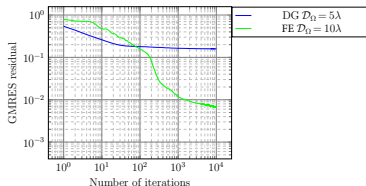
Find $\mathbf{X}_k \in \mathbb{K}_k$ minimizing $\|\mathbf{A}\mathbf{X}_k - \mathbf{F}\|_2$.

If the spectrum of $\mathbf{A}$ lies in $B(\lambda_0, r)$ with $r < |\lambda_0|$, then for all $\varepsilon > 0$

$$\|\mathbf{X}_k - \mathbf{X}\|_2 \leq C_\varepsilon \left(\frac{|\lambda|_{\max} - |\lambda|_{\min}}{|\lambda|_{\max} + |\lambda|_{\min}} + \varepsilon \right)^k \|\mathbf{F}\|_2$$

Efficient implementation by Saad and Schultz^a

- GMRES is **not exact**;
- **GMRES converges very slowly**.



- We therefore use **restarts** to clear the memory.

This is one of the curses of large dimension.

^aY. Saad, M. H. Schultz, GMRES: a generalized minimal residual algorithm for solving nonsymmetric linear systems, SIAM J. Sci. Stat. Comput. 7(3), 856–869, 1986.

Iterative Trefftz methods

- The convergence speed ¹ depends on **matrix spectrum $\mathbf{A}$** .

$$\|[\mathbf{x}]^m - [\mathbf{x}]\| \leq C \|r^0\| \varepsilon^m,$$

with

$$\varepsilon^m = \min_{\substack{p \in \mathcal{P}^m \\ p(0)=1}} \max_{\lambda_i \in \rho(\mathbf{A})} |p(\lambda_i)|.$$

- The ability to handle large cases is governed by the **memory**

$$\text{size}(\text{Krylov}) + \text{nnz}(\mathbf{A}) = N_{\text{kry}} \times N \times N_{\text{elem}} + N^2 \times 7 \times N_{\text{elem}}$$

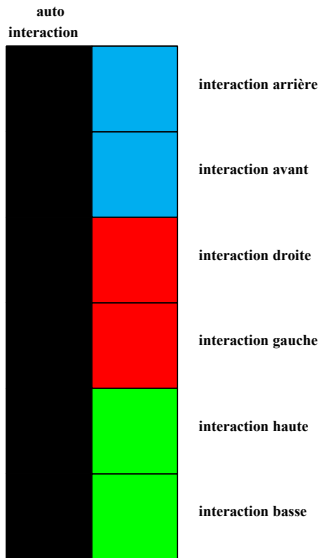
- Convergence speed is governed by eigenvalues close to 0.

We propose strategies to

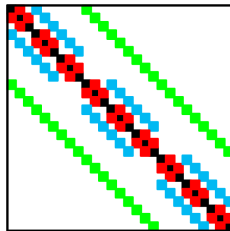
1. **move the spectrum of $\mathbf{A}$ away from 0.**
2. **reduce the memory cost.**

¹Saad, Youcef and Schultz, Martin H, GMRES: A generalized minimal residual algorithm for solving nonsymmetric linear systems, SIAM, 7, 3, 856–869, 1986

Memory cost without assembly $7 \times N^2$



Cartesian mesh $\implies$ structured matrix



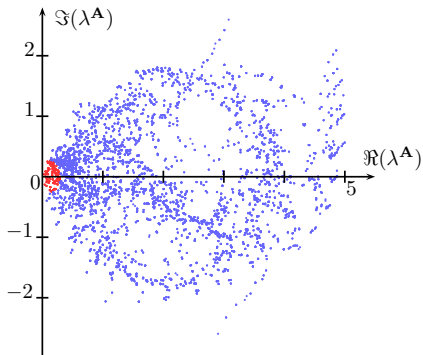
A

Without assembling, Ax can be computed for any x
(the seven blocks: self-interaction, then the back, front, right, left, top and bottom face interactions)

Cessenat-Després preconditioner

The convergence of Krylov method depends on **A spectrum** and the distance between **eigenvalues** and 0.

Idea: **preconditioning** $A^\# A[x] = A^\# F$.

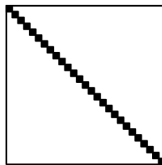


Real and imaginary parts of **A** spectrum, for a domain with size 6λ .

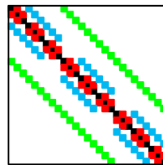
$$A = M + N$$

Cessenat-Després decomposition ^a

M



N



^aO. Cessenat and B. Despres, (1998). Application of an ultra weak variational formulation of elliptic PDEs to the two-dimensional Helmholtz problem. SIAM journal on numerical analysis, 35(1), 255-299.

Cessenat-Després preconditioner

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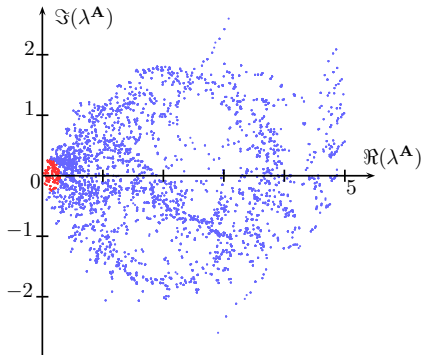
Idea: $\mathbf{A}^\# \mathbf{A}[\mathbf{x}] = \mathbf{A}^\# \mathbf{F}$.

$$\mathbf{A}^\# (\mathbf{M} + \mathbf{N})[\mathbf{x}] = \mathbf{A}^\# \mathbf{F}$$

$$\mathbf{A}^\# \mathbf{M}[\mathbf{x}]^{n+1} = \mathbf{A}^\# \mathbf{N}[\mathbf{x}]^n + \mathbf{A}^\# \mathbf{F}$$

$$\rightarrow \mathbf{A}^\# = \mathbf{M}^{-1}$$

$$\tilde{\mathbf{A}} = \mathbf{M}^{-1} \mathbf{A} = \mathbf{Id} + \mathbf{M}^{-1} \mathbf{N}$$

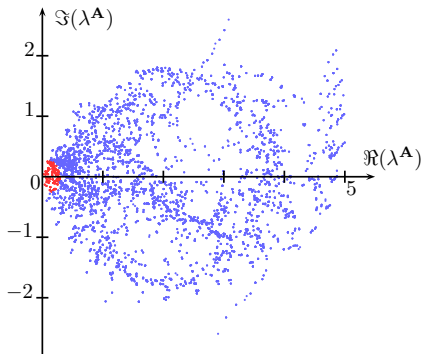


Real and imaginary parts of **A** spectrum, for a domain
with size 6λ .

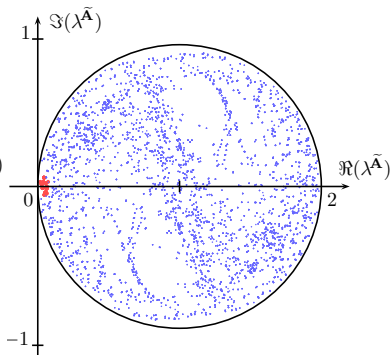
Cessenat-Després preconditioner

The convergence of Krylov method depends on
A spectrum and the distance between **eigenvalues** and 0.

$\tilde{\mathbf{A}} = \mathbf{Id} + \mathbf{M}^{-1}\mathbf{N}$ is a contraction



Real and imaginary parts of $\mathbf{A}$ spectrum, for a domain with size 6λ .



Real and imaginary part of $\tilde{\mathbf{A}}$ spectrum, for a domain size 6λ .

Generalised Trefftz approach: inspiration from the DDM interpretation

- Objective of the Trefftz approach: **enforce the continuity of p and v_N** at the interface between the cells T and K . Technically, it reads (with

$$v_N^T := (i\kappa\mu_T)^{-1}\partial_{\mathbf{n}_T}p^T:$$

$$\begin{cases} p^T - Z_K v_N^T &= p^K + Z_K v_N^K \\ p^K - Z_T v_N^K &= p^T + Z_T v_N^T \end{cases} \iff p^T = p^K \quad \text{and} \quad v_N^T + v_N^K = 0.$$

- DD Methods rather go from the continuity to transmission conditions¹, thanks to **transmission operators** $\mathcal{O}_{T/K,F}$ (generalising the impedances $Z_{T/K}$):

$$p^T = p^K \quad \text{and} \quad v_N^T + v_N^K = 0 \iff \begin{cases} p^T - \mathcal{O}_{T,F}(v_N^T) &= p^K + \mathcal{O}_{T,F}(v_N^K), \\ p^K - \mathcal{O}_{K,F}(v_N^K) &= p^T + \mathcal{O}_{K,F}(v_N^T). \end{cases}$$

- Optimal choice** is the **exterior Dirichlet-to-Neumann operator**!

¹M. Jensen. Discontinuous Galerkin methods for Friedrichs systems with irregular solutions. PhD thesis, University of Oxford, 2004.

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- Optimal** choice is the **exterior Dirichlet-to-Neumann operator**!

$\implies$ Transfer more information at the interface.

$\implies$ Copy this principle of **transmission operators** for the Trefftz method, searched as approximations of the DtN.

¹M. Jensen. Discontinuous Galerkin methods for Friedrichs systems with irregular solutions. (PhD thesis, University of Oxford, 2004)

Generalised Trefftz approach: a class of suitable transmission operators

We look for $\mathcal{T}_{T,F}$ as a DtN approximation:

- Zero-th order approximation¹: $\mathcal{DtN} \approx Y \mathcal{J} \rightarrow$ recovers the classic Trefftz.
- Exact operator for classic outgoing waves with respect to the cell centre $\mathbf{x}_0$:

$$\mathcal{DtN}(p)(\mathbf{x}) \approx i \mathbf{d} \cdot \mathbf{n} \frac{H_1^{(1)}(\kappa r)}{H_0^{(1)}(\kappa r)} p(\mathbf{x}) \text{ (Green kernel),}$$

and

$$\mathcal{DtN}(p)(\mathbf{x}) \approx \mathbf{d}(\mathbf{x}) \cdot \mathbf{n} p(\mathbf{x}) \text{ (directive PWs),}$$

for $\mathbf{d} = (\mathbf{x} - \mathbf{x}_0)/r$ and $r = |\mathbf{x} - \mathbf{x}_0|$.

- Operator with same principal symbol as the classic approximation²

$$\mathcal{DtN} \approx \sqrt{\mathcal{J} + \frac{\Delta_F}{\kappa^2}},$$

and Padé-type approximations³.

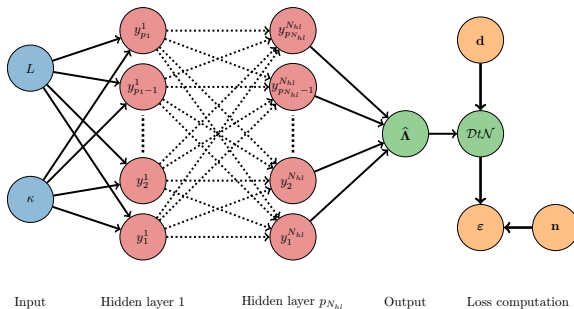
- Data-driven approximation: allows to enforce $\mathcal{T}_{T,F} \in \mathcal{C}_{Trefftz}$.

¹B. Després. "Méthodes de décomposition de domaine pour la propagation d'ondes en régime harmonique. Le théorème de Borg pour l'équation de Hill vectorielle". Thèse de doctorat. Paris 9, Jan. 1991.

²M. El Bouajaji, X. Antoine, C. Geuzaine, 'Approximate local magnetic-to-electric surface operators for time-harmonic Maxwell's equations'. Journal of Computational Physics, 2014.

³B. Després, A. Nicolopoulos, B. Thierry, 'Corners and stable optimized domain decomposition methods for the Helmholtz problem'. Numerische Mathematik, 2021.

A data-driven approximation of the DtN operator: principle



- To assess the inference: exemplary Dirichlet data $\mathbf{d}$.
- Approximation $\mathcal{DtN}(\mathbf{d})$ of the Neumann data.
- Exemplary Neumann data $\mathbf{n}$.
- Error $\boldsymbol{\varepsilon} = \mathbf{n} - \mathcal{DtN}(\mathbf{d})$ we aim at reducing.

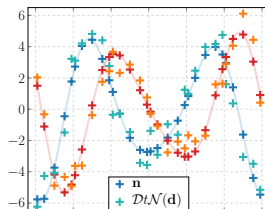
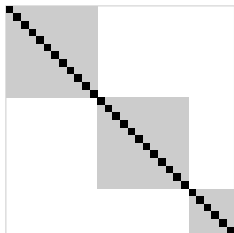


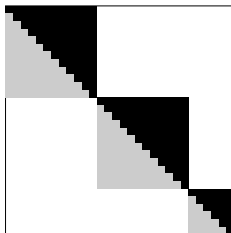
Figure: Error $\boldsymbol{\varepsilon} = \mathbf{n} - \mathcal{DtN}(\mathbf{d})$.

A data-driven approximation of the DtN operator: training

- **Normalised datasets** of 200,000 elements, based on single plane waves, sums of plane waves or a Hankel function.
- Six hidden layers of increasing size, hyperbolic tangent activation.
- **Weighted mean-square error** (with respect to the quadrature), optimised by **stochastic gradient descent**, learning rate $\alpha_{lr} = 0.05$.
- Two shapes of the NN output $\hat{\Lambda}$: **strictly positive diagonal** or **block triangular** — both enforcing $\mathcal{T}_{T,F} \in \mathcal{C}_{Trefftz}$.

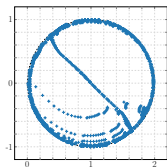


(a) Sparsity pattern of a diagonal $\hat{\Lambda}$

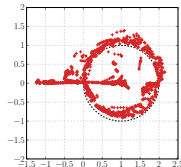


(b) Sparsity pattern of a 'full' $\hat{\Lambda}$

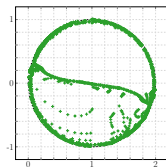
Numerical discussion: eigenvalues distribution



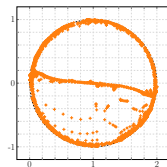
(c) $Trefftz_orig$



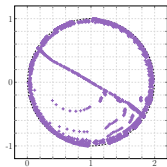
(d) $Trefftz_sqrt$



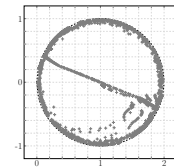
(e) $Trefftz_Padé1$



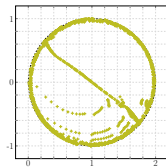
(f) $Trefftz_Padé2$



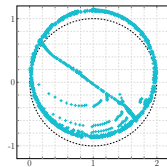
(g) $Trefftz_NN\ Diag$



(h) $Trefftz_NN\ Full$



(i) $Trefftz_PW\ Centre$

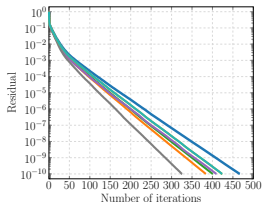


(j) $Trefftz_Green\ Centre$

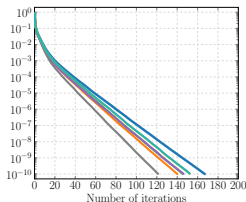
Figure: Eigenvalues distribution of the preconditioned Trefftz matrices for different choices of the DtN approximation.

$\Rightarrow$ Importance to **satisfy all the conditions** of $\mathcal{E}_{Trefftz}$, but insufficient to have an insight on the iterative improvement.

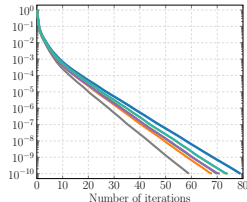
Numerical discussion: improvement of the iterative solution



(a) $h = 3\lambda$ and $N_{Kry} = 20$



(b) $h = 3\lambda$ and $N_{Kry} = 50$



(c) $h = 3\lambda$ and $N_{Kry} = 100$

Figure: Evolution of the residual along restarted GMRES(N_{Kry}) iterations for different choices of the DtN approximation: *Trefftz_orig* (—), *Trefftz_Padé1* (—), *Trefftz_Padé2* (—), *Trefftz_NNDiag* (—), *Trefftz_NNFull* (—), *Trefftz_PWCentre* (—) and *Trefftz_GreenCentre* (—).

- The data-driven approaches offer at most **30% of iteration gain**, while at least **maintaining the original accuracy**.

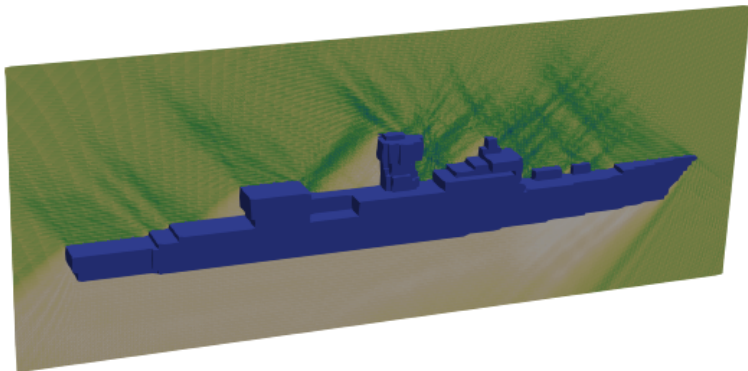
Scaling up

- Result taken from **M. Sirdey** PhD thesis (ONERA-Makutu) co-advised by **S. Pernet** (ONERA)
- Electromagnetic diffraction of a frigate illuminated from above

$$300 \times 100 \times 100 \lambda^3.$$

- Calculations involving more than **a billion unknowns**

Unknown	Matrix	LU Factorization	Method
16 GB	3 TB	1000 TB	800 GB



Conclusion

The **numerical quasi-Trefftz method** builds the discrete space from a **local solution operator** $S_{K,h}$ instead of analytical solutions:

$$X_H = S_K V_H \quad \longrightarrow \quad X_{H,h} = S_{K,h} V_H.$$

- it applies to **any local solver** — finite elements, local boundary elements, spectral differences / flux reconstruction — and therefore to **heterogeneous and complex media**;
- it **controls round-off errors** and reduces the number of unknowns, where plane waves become numerically dependent;
- it is **superconvergent**: the expected order $+\frac{1}{2}$;
- it fits the **hyperbolic / Friedrichs formalism** (energy conservation, upwind flux $F[n]_{\pm}$), which makes the formulation **coercive**;
- the system is then solved **iteratively** (algebraic domain decomposition), and the iterative convergence is improved through **optimized transmission operators** (DtN approximations, including a data-driven one);
- scaling up to **more than a billion unknowns**.

Perspectives

- quasi-Trefftz bases for strongly heterogeneous cells and for convected acoustics;
- coupling several local solvers within a single master Trefftz solver;
- exascale prototyping within the PEPR Numpex.

Some references

- M. Rivet, S. Pernet, S. Tordeux, *Optimised correction polynomial functions for the flux reconstruction method in time-harmonic electromagnetism*, Applied Mathematics Letters, 2024.
- H. Barucq, A. Bendali, J. Diaz, S. Tordeux, *Local strategies for improving the conditioning of the plane-wave Ultra-Weak Variational Formulation*, Journal of Computational Physics, 441, 110449, 2021.
- S. Pernet, M. Sirdey, S. Tordeux, *Ultra-weak variational formulation for heterogeneous Maxwell problem in the context of high performance computing*, ESAIM Proc., 2022.
- H. S. Fure, S. Pernet, M. Sirdey, S. Tordeux, *A discontinuous Galerkin Trefftz type method for solving the two dimensional Maxwell equations*, SN Partial Differential Equations and Applications, 1(23), 19, 2020.
- H. Barucq, A. Bendali, M'B. Fares, V. Mattesi, S. Tordeux, *A symmetric Trefftz-DG formulation based on a local boundary element method for the solution of the Helmholtz equation*, Journal of Computational Physics, 330, 1069–1092, 2017.
- GoTem3: an open-source Fortran 3D Trefftz Maxwell code.

Thank you for your attention.